Didactic Note on Prof. Maday's LSPG-Metric Proposals

Aligned with `manuscript.tex` notation

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1 Purpose of this note

This note rewrites Prof. Maday's proposal using the same symbols as your manuscript:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}\ \overline{\mathbf{V}}], \qquad \mathbf{q}_N = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}, \qquad \tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \overline{\mathbf{V}}\bar{\mathbf{q}}.$$

The goal is to explain, step by step, why small errors in high coefficients can strongly damage low-coefficient recovery in LSPG, and what three concrete remedies are being proposed.

2 Notation dictionary (quick reference)

We keep the notation consistent with `manuscript.tex`. Let N denote full state dimension and N_s the number of snapshots.

Symbol	Meaning
$\boldsymbol{\mu}$	parameter vector (physical/ROM input parameter)
$\mathbf{u} \in \mathbb{R}^N$	full state vector
$\mathbf{u}_{\text{ref}}$	reference state used for affine reconstruction
$\mathbf{V}_{\text{tot}} = [\mathbf{V}\ \overline{\mathbf{V}}]$	reduced basis split into low and high blocks
$\mathbf{q}_N = [\mathbf{q}; \bar{\mathbf{q}}]$	reduced coefficients in full reduced space
$\mathbf{q} \in \mathbb{R}^n$	low/retained coefficients (online unknown in Case 2)
$\bar{\mathbf{q}} \in \mathbb{R}^{\bar{n}}$	high/truncated coefficients (closure output in Case 2)
$\tilde{\mathbf{u}}$	reconstructed state from reduced coordinates
$\mathbf{r}(\mathbf{u}; \boldsymbol{\mu})$	time-discrete HDM residual
$\mathbf{J} = \partial \mathbf{r} / \partial \mathbf{u}$	residual Jacobian w.r.t. state
$\mathbf{P}$	SPD residual metric in LSPG norm $\|\mathbf{r}\|_{\mathbf{P}}^2 = \mathbf{r}^T \mathbf{P} \mathbf{r}$
$\Phi(\mathbf{q}; \bar{\mathbf{q}})$	local reduced LSPG objective (least-squares function)
$\mathbf{S} \in \mathbb{R}^{N \times N_s}$	snapshot matrix (typically centered snapshots as columns)
$\mathbf{W}$	SPD POD metric (state-space inner product matrix)
$\mathbf{C} = \mathbf{S}^T \mathbf{W} \mathbf{S}$	weighted snapshot correlation matrix
$\boldsymbol{\phi}$	POD mode in full state space
$\mathbf{z}$	coefficient vector in snapshot space ($\boldsymbol{\phi} = \mathbf{S}\mathbf{z}$)
λ	eigenvalue associated with POD mode
$\mathbf{A} = \sum_s \omega_s \mathbf{J}_s^T \mathbf{P}_s \mathbf{J}_s$	averaged LSPG sensitivity metric
ω_s	nonnegative sample weights, $\sum_s \omega_s = 1$
ε	small regularization weight in $\mathbf{W}_{\text{eff}}$
$\mathbf{H}_{LL}, \mathbf{H}_{LH}, \mathbf{H}_{HH}, \mathbf{H}_{HL}$	low/high block matrices induced by $\mathbf{A}$
$\mathbf{K}, \mathbf{L}$	correction matrices used to update high/low basis blocks

3 Where the issue comes from (in your current formulation)

In Case 2 of your manuscript, the closure gives

$$\bar{\mathbf{q}} = \mathcal{M}(\boldsymbol{\mu}, t),$$

and the online unknown is only $\mathbf{q}$, obtained by solving an LSPG least-squares problem at each time step.

In practice, this means:

- $\bar{\mathbf{q}}$ is injected from a regressor (ANN, GPR, etc.),
- then LSPG solves for $\mathbf{q}$.

Prof. Maday's key observation is:

- if $\bar{\mathbf{q}}$ is exact, LSPG can recover a good $\mathbf{q}$,
- but if $\bar{\mathbf{q}}$ has even a small error, the solved $\mathbf{q}$ can drift a lot.

So the true problem is not only "predict high coefficients accurately"; it is also "make the reduced basis split robust so that high-mode errors do not contaminate low modes too much".

4 Linearized contamination mechanism: full derivation

Let $\mathbf{r}(\mathbf{u};\boldsymbol{\mu})$ be the time-discrete residual and $\mathbf{J} = \partial\mathbf{r}/\partial\mathbf{u}$ the Jacobian at the current linearization point.

Assume the residual norm is weighted by a symmetric positive matrix $\mathbf{P}$:

$$\|\mathbf{r}\|_{\mathbf{P}}^2 = \mathbf{r}^T\mathbf{P}\,\mathbf{r}.$$

(If Euclidean residual norm is used, $\mathbf{P} = \mathbf{I}$.)

Define the low/high Jacobian blocks

$$\mathbf{J}_L := \mathbf{J}\mathbf{V}, \qquad \mathbf{J}_H := \mathbf{J}\overline{\mathbf{V}}.$$

Step 1: local LSPG objective in low coordinates

For fixed time step and parameter, define

$$\Phi(\mathbf{q};\bar{\mathbf{q}}) := \frac{1}{2}\left\|\mathbf{r}\big(\mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q} + \overline{\mathbf{V}}\bar{\mathbf{q}};\boldsymbol{\mu}\big)\right\|_{\mathbf{P}}^2.$$

Here Φ is the local least-squares objective seen by the low coordinates $\mathbf{q}$ when the high coordinates $\bar{\mathbf{q}}$ are injected/fixed by closure. At a local optimum $(\mathbf{q},\bar{\mathbf{q}})$, first-order optimality with respect to $\mathbf{q}$ gives

$$\nabla_{\mathbf{q}}\Phi = \mathbf{J}_L^T\mathbf{P}\,\mathbf{r} = \mathbf{0}.$$

Step 2: perturb only the high block

Now perturb the high block:

$$\bar{\mathbf{q}} \mapsto \bar{\mathbf{q}} + \delta\bar{\mathbf{q}},$$

and let the low solution shift to $\mathbf{q} + \delta\mathbf{q}$. Linearizing the residual:

$$\mathbf{r}(\mathbf{q}+\delta\mathbf{q},\bar{\mathbf{q}}+\delta\bar{\mathbf{q}}) \approx \mathbf{r} + \mathbf{J}_L\,\delta\mathbf{q} + \mathbf{J}_H\,\delta\bar{\mathbf{q}}.$$

Step 3: linearized least-squares normal equations

With fixed $\delta\bar{\mathbf{q}}$, the local problem in $\delta\mathbf{q}$ is

$$\min_{\delta\mathbf{q}} \frac{1}{2} \left\| \mathbf{r} + \mathbf{J}_L \delta\mathbf{q} + \mathbf{J}_H \delta\bar{\mathbf{q}} \right\|_{\mathbf{P}}^2 .$$

Normal equations:

$$\mathbf{J}_L^T \mathbf{P} \left(\mathbf{r} + \mathbf{J}_L \delta\mathbf{q} + \mathbf{J}_H \delta\bar{\mathbf{q}} \right) = \mathbf{0}.$$

Subtracting the unperturbed stationarity condition $\mathbf{J}_L^T \mathbf{P} \mathbf{r} = \mathbf{0}$:

$$\mathbf{J}_L^T \mathbf{P} \mathbf{J}_L \, \delta\mathbf{q} + \mathbf{J}_L^T \mathbf{P} \mathbf{J}_H \, \delta\bar{\mathbf{q}} = \mathbf{0}.$$

If $\mathbf{J}_L^T \mathbf{P} \mathbf{J}_L$ is invertible, then

$$\delta\mathbf{q} = - \left(\mathbf{J}_L^T \mathbf{P} \mathbf{J}_L \right)^{-1} \mathbf{J}_L^T \mathbf{P} \mathbf{J}_H \, \delta\bar{\mathbf{q}}.$$

Step 4: transfer operator

Define

$$\mathbf{T} := \left(\mathbf{J}_L^T \mathbf{P} \mathbf{J}_L \right)^{-1} \mathbf{J}_L^T \mathbf{P} \mathbf{J}_H.$$

Then

$$\delta\mathbf{q} = -\mathbf{T} \, \delta\bar{\mathbf{q}}.$$

So if $\|\mathbf{T}\|$ is large, small high-block errors can generate large low-block errors.

Plain-language interpretation. This is exactly your symptom: even when the high map looks "reasonable" on some metric, the online LSPG solve can still become fragile because the low/high decomposition is not well decoupled in the *LSPG sensitivity metric*.

5 Proposal 1: use a better POD metric

5.1 Weighted POD baseline

Instead of Euclidean SVD, use weighted POD with SPD matrix $\mathbf{W}$:

$$\mathbf{C} = \mathbf{S}^T \mathbf{W} \mathbf{S}.$$

Here $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_{N_s}] \in \mathbb{R}^{N \times N_s}$ is the snapshot matrix (usually centered snapshots), and $\mathbf{W} \in \mathbb{R}^{N \times N}$ defines the state-space inner product.

Where this comes from (derivation). For one POD mode $\boldsymbol{\phi}$, weighted POD maximizes captured weighted snapshot energy:

$$\max_{\boldsymbol{\phi} \neq \mathbf{0}} \frac{\boldsymbol{\phi}^T \mathbf{W} \mathbf{S} \mathbf{S}^T \mathbf{W} \boldsymbol{\phi}}{\boldsymbol{\phi}^T \mathbf{W} \boldsymbol{\phi}}.$$

The associated stationarity condition is

$$\mathbf{S} \mathbf{S}^T \mathbf{W} \boldsymbol{\phi} = \lambda \boldsymbol{\phi}.$$

Using method of snapshots, write $\boldsymbol{\phi} = \mathbf{S} \mathbf{z}$ (mode in the snapshot span). Substituting:

$$\mathbf{S}^T \mathbf{W} \mathbf{S} \, \mathbf{z} = \lambda \mathbf{z} \quad \Longleftrightarrow \quad \mathbf{C} \mathbf{z} = \lambda \mathbf{z}.$$

Then the weighted POD mode is reconstructed by

$$\boldsymbol{\phi} = \frac{1}{\sqrt{\lambda}}\mathbf{S}\mathbf{z},$$

with normalization $\boldsymbol{\phi}^T\mathbf{W}\boldsymbol{\phi} = 1$.

For your two-component Burgers state, Prof. Maday suggests the natural state metric

$$\mathbf{W} = \mathbf{M}_{\text{block}} = \begin{bmatrix} \mathbf{M}_h & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_h \end{bmatrix},$$

which corresponds to a discrete L^2-type norm of (u_x, u_y) instead of a plain coordinate-vector norm.

5.2 LSPG-aware enriched metric

To directly encode LSPG sensitivity into basis extraction, use

$$\mathbf{W}_{\text{eff}} = \varepsilon\,\mathbf{M}_{\text{block}} \;+\; \sum_s \omega_s\,\mathbf{J}_s^T\mathbf{P}_s\mathbf{J}_s, \qquad \omega_s \geq 0,\; \sum_s \omega_s = 1.$$

Here each sample s corresponds to a representative time-parameter point.

Where this comes from (derivation). Around a sample s, for a small state perturbation $\delta\mathbf{u}$, residual linearization gives

$$\delta\mathbf{r}_s \approx \mathbf{J}_s\,\delta\mathbf{u}.$$

Hence the weighted residual change is

$$\|\delta\mathbf{r}_s\|_{\mathbf{P}_s}^2 \approx \delta\mathbf{u}^T\mathbf{J}_s^T\mathbf{P}_s\mathbf{J}_s\,\delta\mathbf{u}.$$

So $\mathbf{J}_s^T\mathbf{P}_s\mathbf{J}_s$ is the local quadratic form measuring "LSPG sensitivity" to perturbations. A weighted average over representative samples yields

$$\mathbf{A} := \sum_s \omega_s\,\mathbf{J}_s^T\mathbf{P}_s\mathbf{J}_s.$$

Adding $\varepsilon\mathbf{M}_{\text{block}}$ keeps a baseline state metric and regularization:

$$\mathbf{W}_{\text{eff}} = \varepsilon\mathbf{M}_{\text{block}} + \mathbf{A}.$$

Plain-language interpretation. The second term penalizes directions that strongly affect residual minimization, so the basis is adapted to what LSPG actually "cares about", not only to snapshot energy.

6 Proposal 2: modify only high modes (keep V fixed)

This is the proposal Prof. Maday emphasized as attractive because online dimension and online solve structure stay unchanged.

Define averaged LSPG metric

$$\mathbf{A} := \sum_s \omega_s\,\mathbf{J}_s^T\mathbf{P}_s\mathbf{J}_s.$$

Then define blocks

$$\mathbf{H}_{LL} := \mathbf{V}^T\mathbf{A}\mathbf{V}, \qquad \mathbf{H}_{LH} := \mathbf{V}^T\mathbf{A}\overline{\mathbf{V}}.$$

Derivation of the correction formula. Seek a corrected high basis in the form

$$\widetilde{\overline{\mathbf{V}}} = \overline{\mathbf{V}} - \mathbf{V}\mathbf{K},$$

where $\mathbf{K}$ is unknown. Impose metric-orthogonality against low modes:

$$\mathbf{V}^T\mathbf{A}\widetilde{\overline{\mathbf{V}}} = \mathbf{0}.$$

Substitute:

$$\mathbf{V}^T\mathbf{A}(\overline{\mathbf{V}} - \mathbf{V}\mathbf{K}) = \mathbf{H}_{LH} - \mathbf{H}_{LL}\mathbf{K} = \mathbf{0}.$$

So

$$\mathbf{H}_{LL}\mathbf{K} = \mathbf{H}_{LH}.$$

If $\mathbf{H}_{LL}$ is invertible,

$$\mathbf{K} = \mathbf{H}_{LL}^{-1}\mathbf{H}_{LH}.$$

Hence

$$\widetilde{\overline{\mathbf{V}}} = \overline{\mathbf{V}} - \mathbf{V}\,\mathbf{H}_{LL}^{-1}\mathbf{H}_{LH}.$$

Direct verification.

$$\mathbf{V}^T\mathbf{A}\widetilde{\overline{\mathbf{V}}} = \mathbf{H}_{LH} - \mathbf{H}_{LL}\mathbf{H}_{LL}^{-1}\mathbf{H}_{LH} = \mathbf{0}.$$

What this buys you. Low modes are untouched, high modes are "orthogonalized" against low modes in the averaged LSPG metric, and the low/high contamination channel is reduced.

Coordinate mapping note (important). Let

$$\mathbf{K} := \mathbf{H}_{LL}^{-1}\mathbf{H}_{LH}.$$

Because $\widetilde{\overline{\mathbf{V}}} = \overline{\mathbf{V}} - \mathbf{V}\mathbf{K}$, the same state can be written as

$$\mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \overline{\mathbf{V}}\bar{\mathbf{q}} = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \left(\widetilde{\overline{\mathbf{V}}} + \mathbf{V}\mathbf{K}\right)\bar{\mathbf{q}} = \mathbf{u}_{\text{ref}} + \mathbf{V}\left(\mathbf{q} + \mathbf{K}\bar{\mathbf{q}}\right) + \widetilde{\overline{\mathbf{V}}}\,\bar{\mathbf{q}}.$$

with one convenient choice

$$\widetilde{\bar{\mathbf{q}}} = \bar{\mathbf{q}}, \qquad \widetilde{\mathbf{q}} = \mathbf{q} + \mathbf{K}\bar{\mathbf{q}}.$$

So the "high" coordinates can be kept as-is, while low coordinates are shifted consistently.

7 Proposal 3: modify only low modes (keep $\overline{\mathbf{V}}$ fixed)

Define

$$\mathbf{H}_{HH} := \overline{\mathbf{V}}^T\mathbf{A}\overline{\mathbf{V}}, \qquad \mathbf{H}_{HL} := \overline{\mathbf{V}}^T\mathbf{A}\mathbf{V}.$$

Derivation of the correction formula. Keep $\overline{\mathbf{V}}$ fixed and seek

$$\widetilde{\mathbf{V}} = \mathbf{V} - \overline{\mathbf{V}}\mathbf{L}$$

such that

$$\overline{\mathbf{V}}^T\mathbf{A}\widetilde{\mathbf{V}} = \mathbf{0}.$$

Substitute:

$$\overline{\mathbf{V}}^T\mathbf{A}(\mathbf{V} - \overline{\mathbf{V}}\mathbf{L}) = \mathbf{H}_{HL} - \mathbf{H}_{HH}\mathbf{L} = \mathbf{0}.$$

Thus

$$\mathbf{H}_{HH}\mathbf{L} = \mathbf{H}_{HL}, \qquad \mathbf{L} = \mathbf{H}_{HH}^{-1}\mathbf{H}_{HL},$$

and therefore

$$\widetilde{\mathbf{V}} = \mathbf{V} - \overline{\mathbf{V}}\,\mathbf{H}_{HH}^{-1}\mathbf{H}_{HL}.$$

Direct verification.
$$\overline{\mathbf{V}}^T \mathbf{A} \widetilde{\mathbf{V}} = \mathbf{H}_{HL} - \mathbf{H}_{HH}\mathbf{H}_{HH}^{-1}\mathbf{H}_{HL} = \mathbf{0}.$$

Interpretation. This is the symmetric version: keep high block fixed, orthogonalize low block in the averaged LSPG metric.

8 How this maps to your current pipeline

Step A: build A on representative samples

Use a subset of Stage-2 trajectories (time and parameter points), compute Jacobians $\mathbf{J}_s$, and assemble
$$\mathbf{A} \approx \sum_s \omega_s \mathbf{J}_s^T \mathbf{P}_s \mathbf{J}_s.$$

Step B: choose one basis intervention

- **Proposal 2 (high-only):** usually the least intrusive for Case 2, because low online unknown size remains n, and you can preserve interpretation of injected high block.
- **Proposal 3 (low-only):** also valid, but often more invasive for current closure workflows.
- **Proposal 1 (metric POD):** can be done independently or combined with Proposal 2/3.

Step C: regenerate consistent reduced data

After basis modification, reduced coefficients should be made consistent with the new basis (either via exact coordinate transformation when available, or by recomputing projection/reduced trajectories).

Step D: retrain closure map and re-run offline/online tests

Then compare:

- low/high coefficient errors,
- LSPG residual norms,
- robustness at off-grid $\boldsymbol{\mu}$ points.

9 What Prof. Maday is really suggesting (one-sentence summary)

Do not treat high-mode prediction error only as a regression problem; redesign the *basis split and metric* so that LSPG dynamics is less sensitive to high-block perturbations.

10 Practical recommendation for your immediate next test

For your current Case-2 setup, the most direct first experiment is:

1. keep n fixed and keep online algorithm unchanged,
2. compute $\mathbf{A}$ from a representative set,
3. apply Proposal 2 (high-mode correction),

4. run the same Stage-2/Stage-3/evaluation protocol and compare against your current baseline.

This isolates whether reduced-basis decoupling improves robustness without changing the online model size.